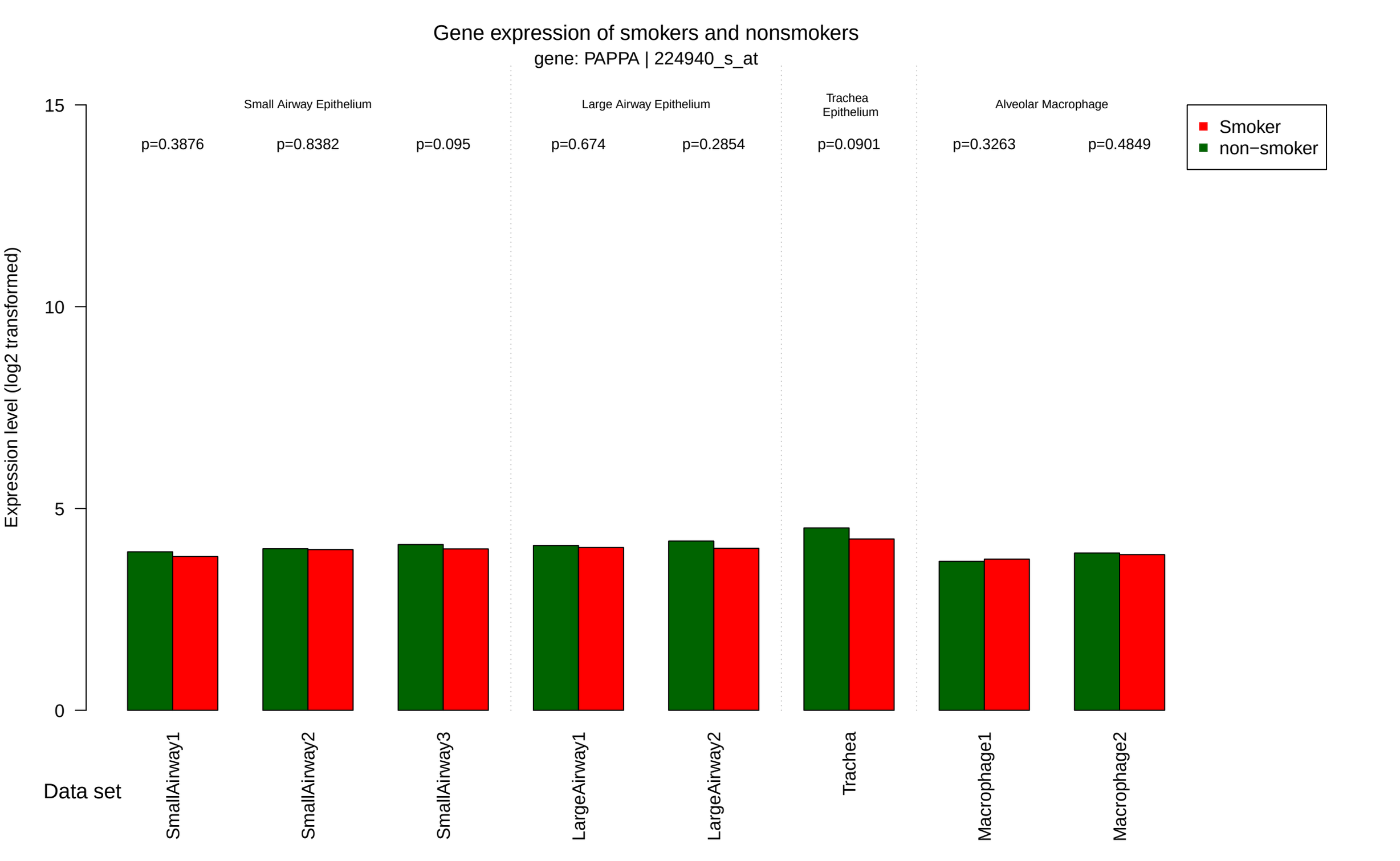
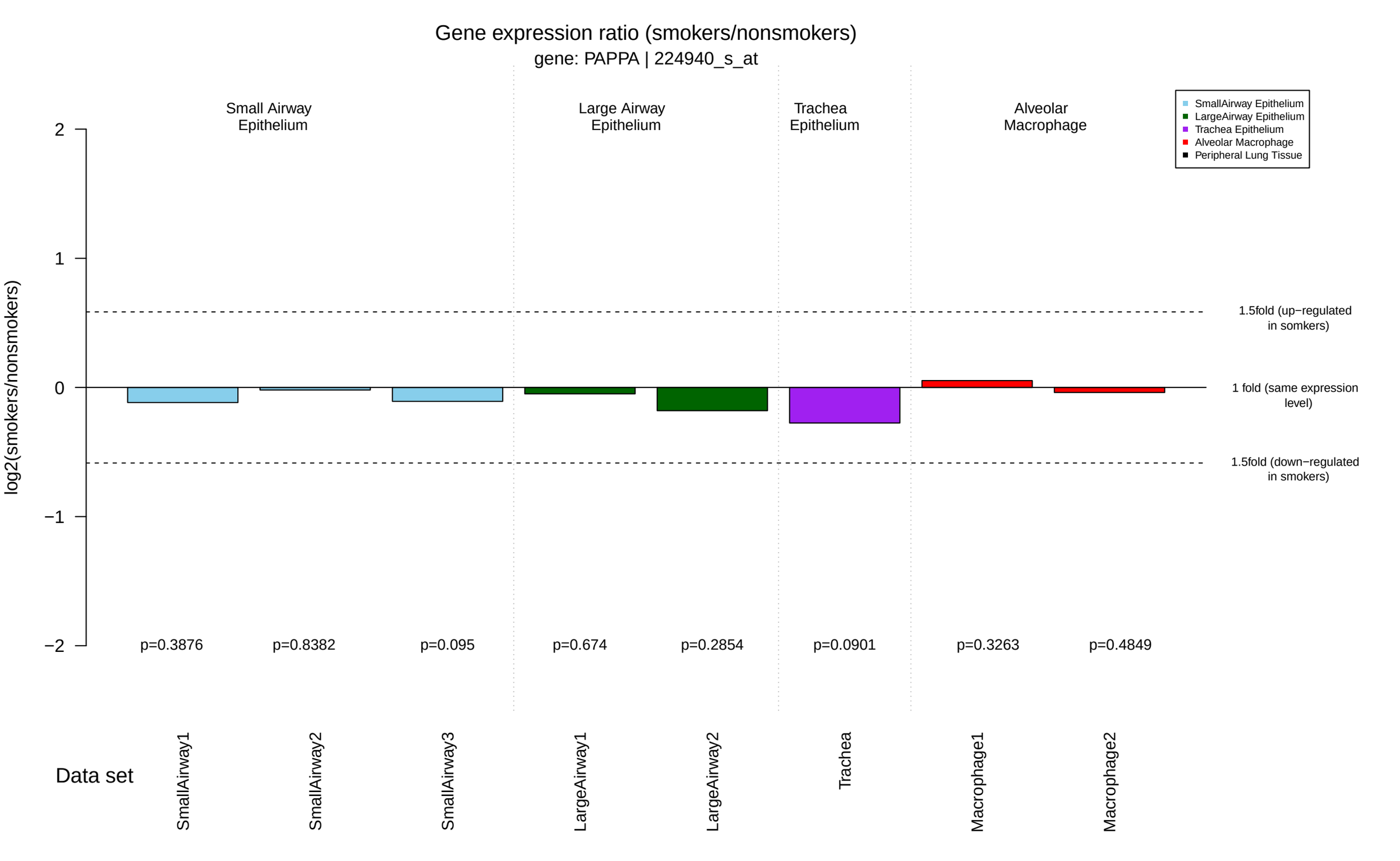
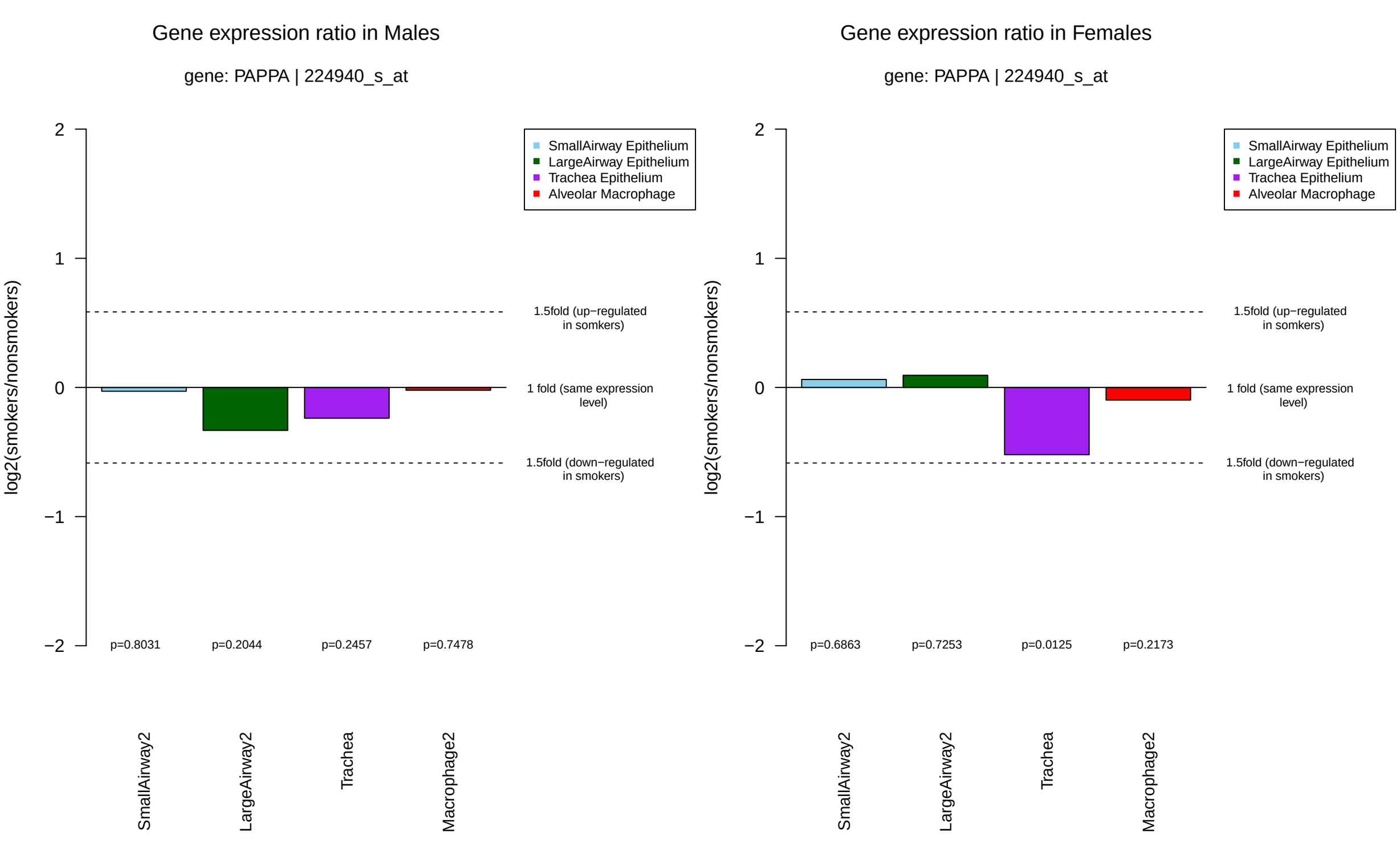
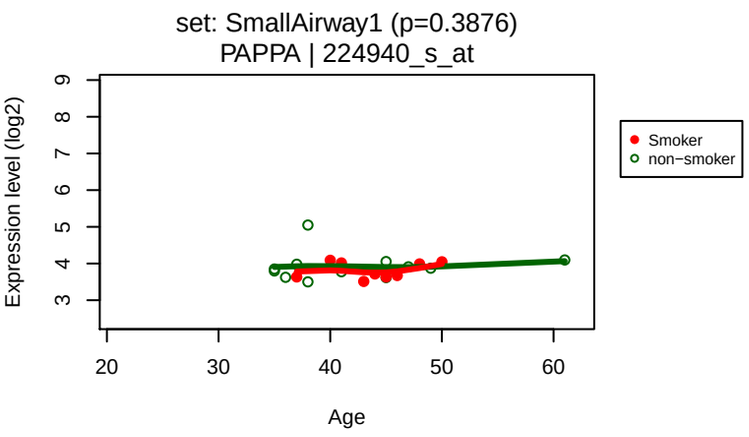
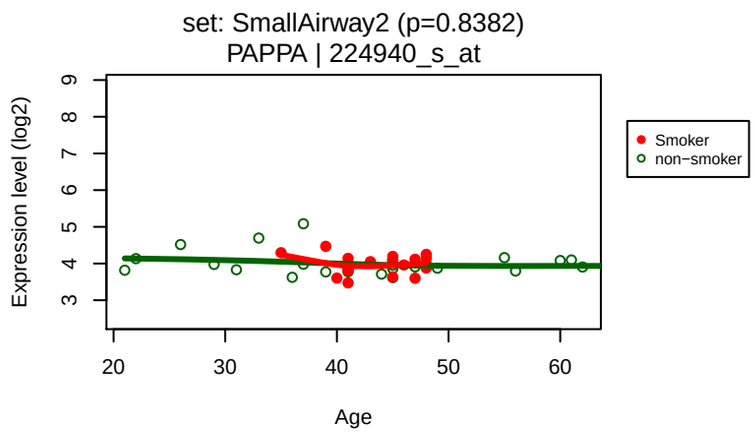
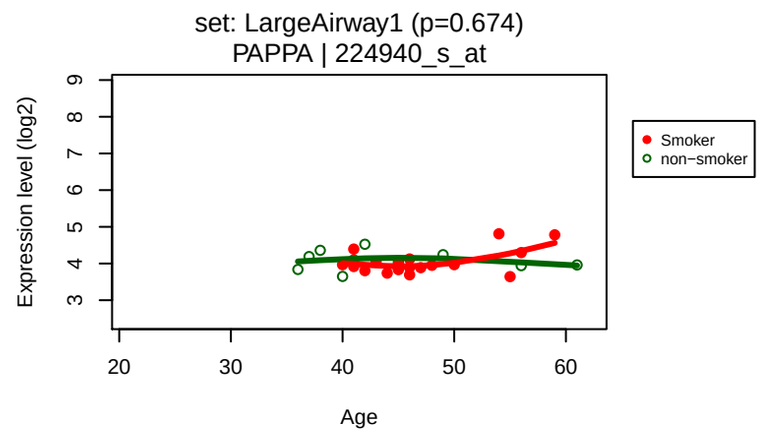

Gene expression of smokers and nonsmokers
gene: PAPPA | 224940_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3876
p=0.8382
p=0.095
p=0.674
p=0.2854
p=0.0901
p=0.3263
p=0.4849
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

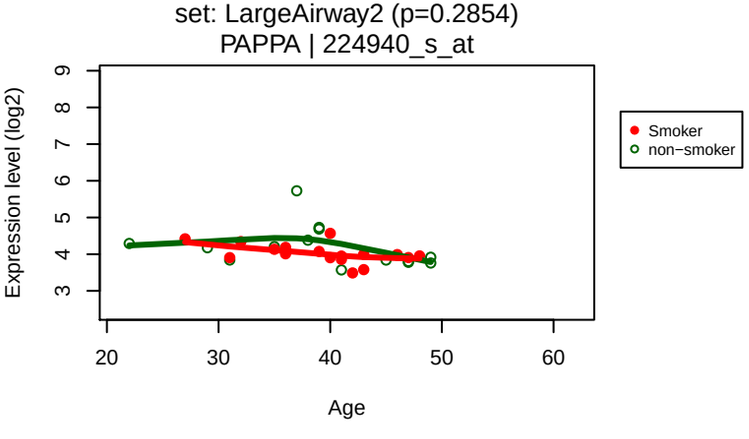

Gene expression ratio (smokers/nonsmokers)
gene: PAPPA | 224940_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3876
p=0.8382
p=0.095
p=0.674
p=0.2854
p=0.0901
p=0.3263
p=0.4849
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

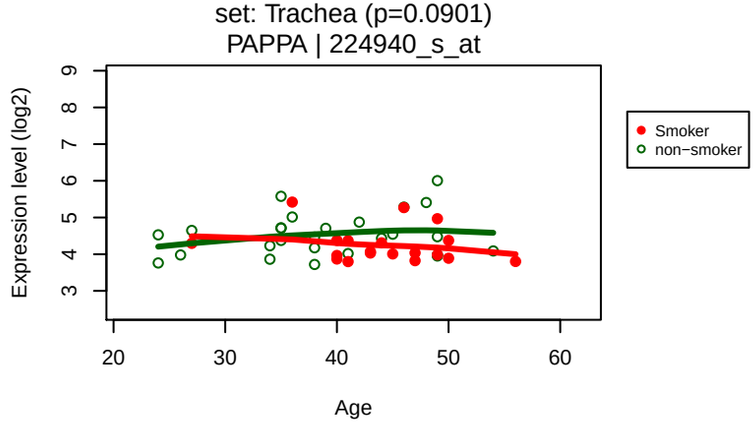

Gene expression ratio in Males
gene: PAPPA | 224940_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8031
p=0.2044
p=0.2457
p=0.7478
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAPPA | 224940_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6863
p=0.7253
p=0.0125
p=0.2173
SmallAirway2
LargeAirway2
Trachea
Macrophage2

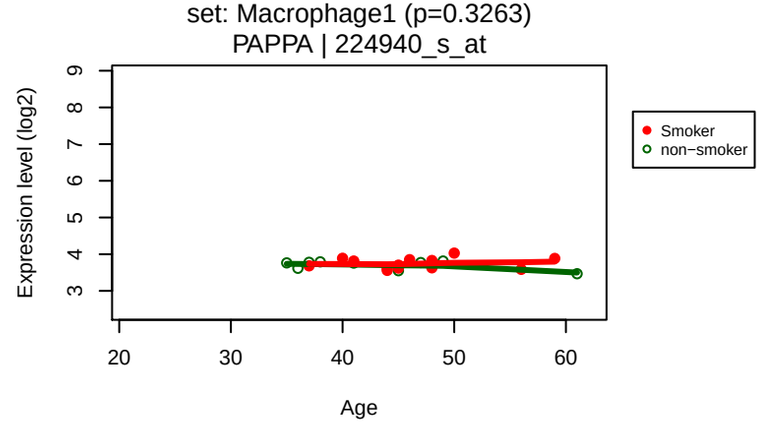

set: SmallAirway1 (p=0.3876)
PAPPA | 224940_s_at
Expression level (log2)
Age
Smoker
non−smoker

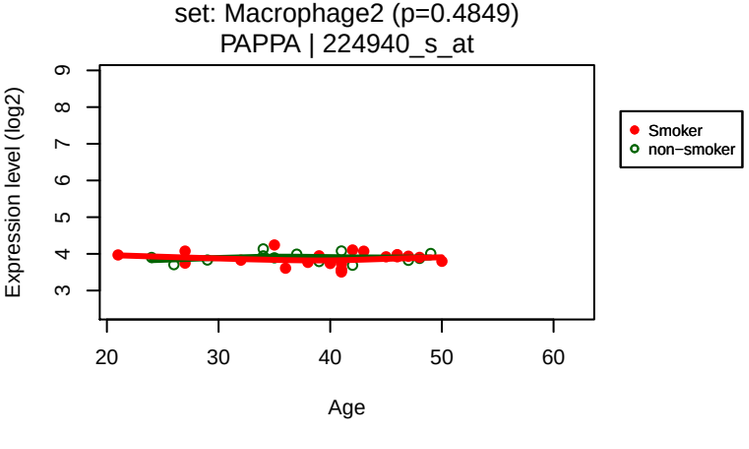

set: SmallAirway2 (p=0.8382)
PAPPA | 224940_s_at
Expression level (log2)
Age
Smoker
non−smoker

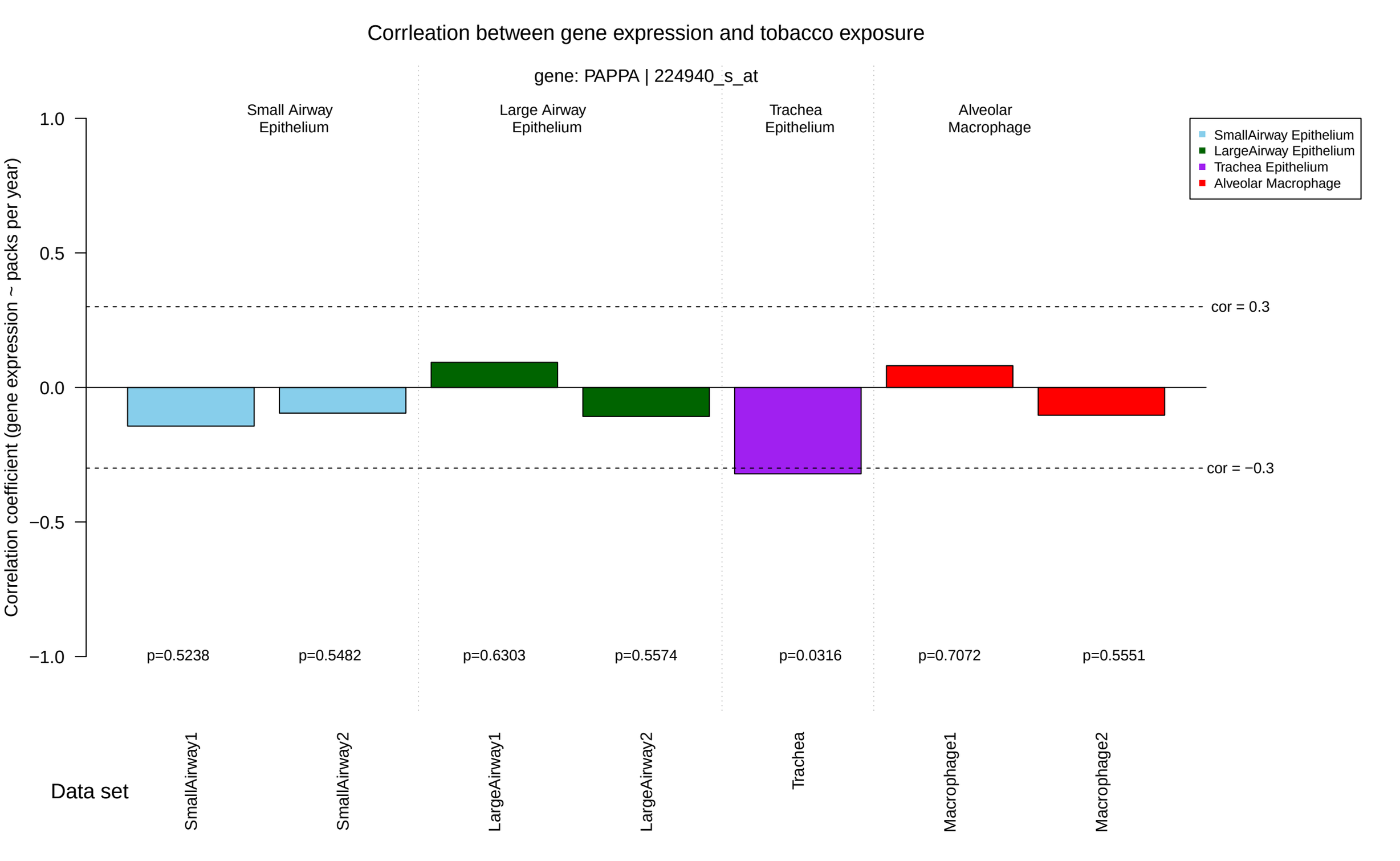

set: LargeAirway1 (p=0.674)
PAPPA | 224940_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2854)
PAPPA | 224940_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0901)
PAPPA | 224940_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3263)
PAPPA | 224940_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4849)
PAPPA | 224940_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAPPA | 224940_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5238
p=0.5482
p=0.6303
p=0.5574
p=0.0316
p=0.7072
p=0.5551
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2